Muhammet Celal Gokalp

ADAS & Automated Driving Engineer | Validation, Simulation & Customer Integration

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PROFILE

Systems and application engineer with nearly eight years in ADAS and automated driving. My experience combines feature ownership and vehicle testing at Jaguar Land Rover through AVL with simulation-based validation at FiveAI / Bosch. I turn requirements and test protocols into test cases, investigate failures, and work with customers, suppliers and software teams to resolve issues. I have also led a small technical team and built practical tools using AI-assisted coding.

EXPERIENCE

Lead Application Engineer (Solutions & Integration)

FiveAI / Bosch | Cambridge, UK | Oct 2021 - Present

- Built more than 120 parameterised OpenSCENARIO 2 scenarios generating thousands of test cases for automated driving validation, including Euro NCAP junction tests and UN R79 automatic lane change tests.
- Defined pass/fail checks and performance criteria, including acceleration and deceleration comfort limits, and evaluated results across more than 1,000 hours of test data.
- Trained Volkswagen Group partner engineers to use FiveAI's simulation tools on the ADA project. Helped establish the Simulation EcoSystem solutions team, investigating issues across simulation tools, maps and partner toolchains.
- Led five engineers for one year on L4 scenario development. Taught the tools and evaluation methods, and divided work between scenario creation and evaluation. This was a technical lead role, not line management.
- Used AI-assisted Python coding to build a JSON test-parameter GUI, a results uploader and data pipelines feeding results into dashboards. Worked in the project monorepo, raised pull requests, and ran simulations locally in a dev container and through the existing CI pipeline.
- Used dt_flow and dt_launcher to run simulations and diagnose issues, and Foxglove to visualise results. Familiar with XODR and OSM map formats and able to read them.
- Built a GitHub Copilot and Copilot CLI helper that creates scenarios from reusable blocks using regulation and Euro NCAP rules as context. The helper was adopted across the project. Taught colleagues to check AI-generated output.

ADAS Systems Engineer (Feature Owner)

AVL Powertrain UK | Coventry, UK | Nov 2018 - Oct 2021

- Worked on-site at Jaguar Land Rover for two years as feature owner for Rear Cross Traffic Alert, Blind Spot Monitor and Rear Pre-Crash. Acted as the technical link between JLR, AVL and the radar supplier.
- Wrote use cases, system state diagrams and requirements in DOORS and MKS Integrity. Broke high-level requirements into lower-level ones, defined module input/output signals and traced test cases to system states.
- Tested features on HiL and ViL rigs, in vehicles and on the proving ground. Analysed CAN logs using Vector CANoe and CANape and tracked defects in RTC / RQM.

EXPERIENCE (CONTINUED)

ADAS Systems Engineer (Feature Owner) - continued

AVL Powertrain UK | Nov 2018 - Oct 2021

- Investigated a late warning at the edge of radar coverage, traced it to the radar's field of view and agreed a fix with the supplier.
- Calibrated a MATLAB / Simulink and IPG CarMaker vehicle model against real recordings to compare vehicle, dyno and simulation testing. Ran parameter sweeps on steering torque PID gains.
- Wrote requirements for ACC, Lane Keep Assist, Highway Assist and Lane Change Assist, including ISO 26262 requirements for a lane change feature. Trained in ISO 26262 and SOTIF.
- At the LEVC factory, checked returned faulty electric machines with a multimeter, collected logger data and passed findings to developers and suppliers for further investigation.

EARLIER EXPERIENCE

Part-time Teacher and Programme Planner

Space Camp Turkey | Izmir, Turkey | Dec 2013 - Jan 2017

- Taught science and space technology to students aged 9 to 16 and planned schedules for student groups and teachers.

Mechanical Engineering Intern

TEMSA Global | Adana, Turkey | May 2014 - Aug 2014

- Calculated power and torque requirements for an electric bus powertrain using MATLAB / Simulink.

EDUCATION

MSc Sustainable Automotive Engineering - Merit

WMG, University of Warwick | Oct 2017 - Sep 2018

- Dissertation: lab testing and thermal modelling of an electric motor stator using MATLAB / Simulink, MotorCAD and Portunus.

BSc Mechanical Engineering

Ege University | Aug 2011 - Feb 2017

- Dissertation: designed, built and programmed a 6-DOF Stewart motion platform using SolidWorks, MATLAB state-space modelling and Arduino control.

TECHNICAL SKILLS

- **FiveAI / Bosch:** OpenSCENARIO 2, simulation-based testing, dt_flow and dt_launcher use and diagnosis, Foxglove result visualisation; familiarity with reading XODR and OSM maps.
- **AVL:** Hardware-in-the-Loop (HiL), Vehicle-in-the-Loop (ViL), vehicle and proving ground testing, performance analysis and root cause investigation; requirements, state diagrams, signal lists and traceability; ASPICE V-model, DOORS, MKS Integrity, RTC / RQM, MATLAB / Simulink, IPG CarMaker and Vector CANoe / CANape.
- **Software and development:** Python for scripts, tools and data pipelines using AI-assisted coding; C++ reading; Git, pull requests, dev containers, existing CI test pipelines, GitHub Copilot and Copilot CLI.
- **Workstation:** Linux as my main workstation for five years; VS Code.

ADDITIONAL INFORMATION

British citizen with a full UK driving licence; willing to relocate within the UK. English (C2) and Turkish (native). Outside work, I repair my own cars and build desktop PCs.